simon-5501-09-slides

2026-10-20

Topics to be covered

- What you will learn
  - Restructuring your data
  - One-tailed tests
  - Checking assumptions
  - Alternative tests
  - R code for alternative tests
  - Paired data
  - R code for paired data
  - Your homework

Two different data structures

- Wide format (two columns)
  - Group 1 in first column
  - Group 2 in second column
- Long format (one column)
  - Group 1 in first n_1 rows of column
  - Group 2 in next n_2 rows of column
  - Could be different order or mixed
  - Additional column for group code

Speaker notes

There are two different data structures that you will encounter in research settings. The wide format places the data from each group in a different column. The long format places both groups in the same column, usually with the first group on the top and the second group on the bottom. It could be reversed, however, or it could be all jumbled together.

You will find that certain analyses are only available in the long format. Other analyses are only available in the wide format.

Being able to move from the wide format to the long format or vice versa is an important programming skill that you should learn early. I don’t want to emphasize programming too much in this class, but I am making an exception here.

Hypothetical data in the wide format

```
1 wide_form
```

```
# A tibble: 4 × 2
    trt   pbo
  <dbl> <dbl>
1    15    34
2    13    31
3    18    36
4    19    NA
```

Speaker notes

Here is an example of data in the wide format. The first column shows the four observations associated with the treatment group and the second column shows the three observations associated with the control group. Notice that we have to pad the shorter column with a missing value code.

Pivoting from wide form to long form

```r
wide_form |>
  pivot_longer(
    cols=c("trt", "pbo"),
    names_to="intervention",
    values_to="outcome") |>
  filter(!is.na(outcome))
```

```
# A tibble: 7 × 2
  intervention outcome
  <chr>           <dbl>
1 trt                15
2 pbo                34
3 trt                13
4 pbo                31
5 trt                18
6 pbo                36
7 trt                19
```

Speaker notes

Here is the code to convert from a wide format to a long format. It uses the pivot_longer function, which is part of the tidyr package which is included in the tidyverse library. Provide the names of columns that you want to pivot. Here there are two columns, “trt” and “pbo”, but you can use pivot_longer with more complex settings with three or more columns.

In the long format, you will need a column that tells you which rows are associated with “trt” and which rows are associated with “pbo”. You specify that with the names_to argument. You specify the column containing the actual numbers using the values+to argument.

Once you have all the data in a single column, you no longer need the missing value placeholder. Get rid of it using the filter command.

Hypothetical data in the long format

```
1 long_form
```

```
# A tibble: 8 × 3
     id result exposed
  <int>  <dbl> <chr>
1     1     45 n
2     2     43 n
3     3     48 n
4     4     49 n
5     1     64 y
6     2     61 y
7     3     66 y
8     4     62 y
```

Speaker notes

This is a different hypothetical dataset with four subjects and measurements taken on each subject when they are exposed to some unspecified toxin and when they are not exposed to that toxin.

This data is arranged in the long format. All of the outcomes are in a single column. The data is paired with the first and fifth rows representing measurements from the first subject (id=1) with the first row representing the outcome for that patient when there was no exposure and the fifth row representing the outcome for that patient when there was exposure to the unspecified toxin. The second and sixth rows represent measurements from the second subject (id=2) with the second row representing the outcome for that patient when there was no exposure and the sixth row representing the the outcome for that patient when there was exposure to the unspecified toxin.

The third and seventh rows represent the third subject. The fourth and eighth rows represent the fourth subject.

Pivoting from long form to wide form

```
long_form |>
  pivot_wider(
    names_from=exposed,
    values_from=result)
```

```
# A tibble: 4 × 3
     id     n     y
  <int> <dbl> <dbl>
1     1    45    64
2     2    43    61
3     3    48    66
4     4    49    62
```

Speaker notes

The pivot_wider function, also part of the tdyr package and included when you load the tidyverse library, restructures the data. You need two columns for the outcome rather than one. The names of those two columns come from the variable “exposed” and the values come from the variable “result”.

Break #1

- What you have learned
  - Restructuring your data
- What’s coming next
  - One-tailed tests

One-tailed tests, 1

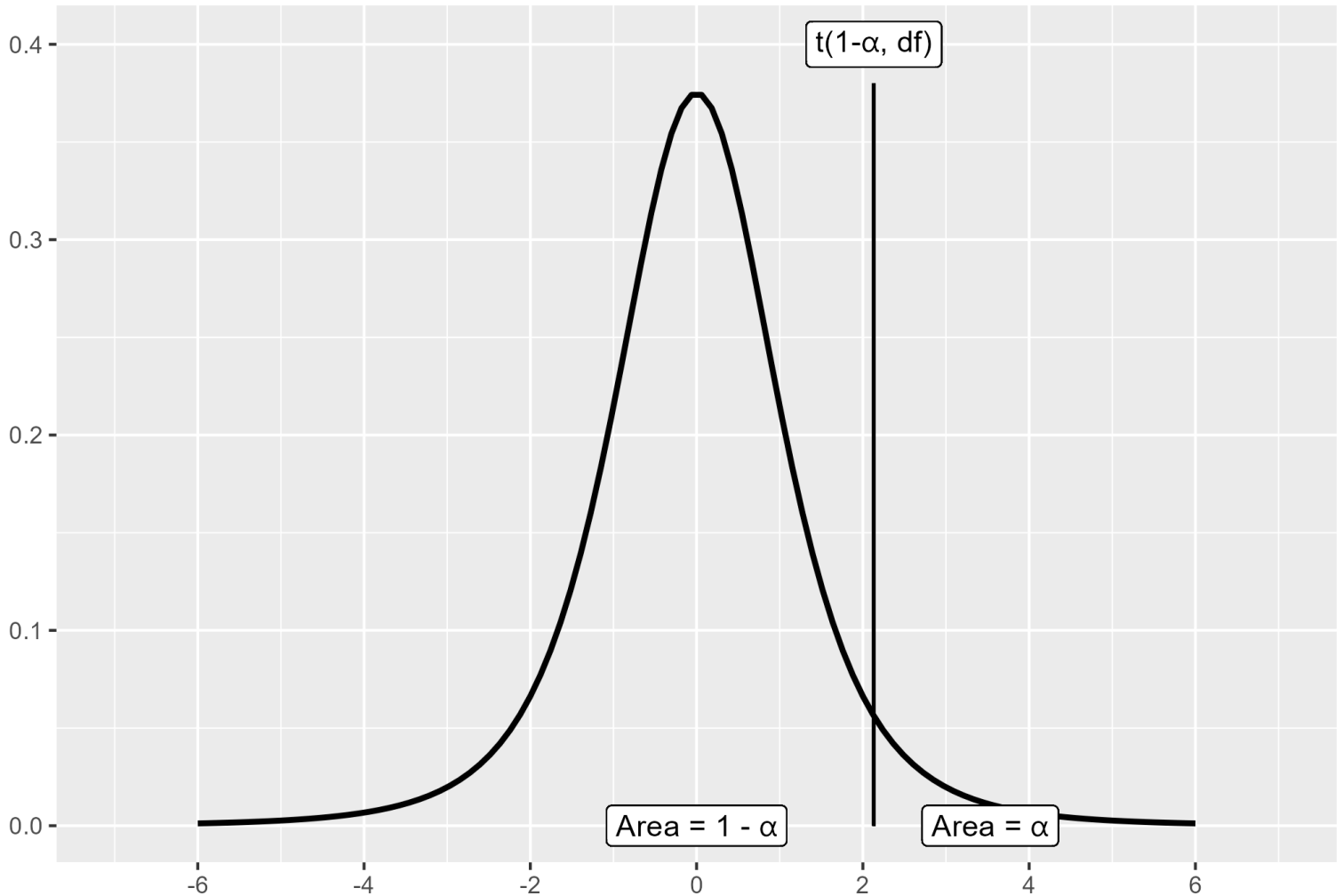
- $H_0: \ \mu_1 - \mu_2 \leq 0$
- $H_1: \ \mu_1 - \mu_2 > 0$
  - Accept H_0 if $\bar{X}_1 - \bar{X}_2$ is large negative
  - Accept H_0 if $\bar{X}_1 - \bar{X}_2$ is close to zero
  - Reject H_0 if $\bar{X}_1 - \bar{X}_2$ is large positive
- More precisely,
  - Accept H_0 if $T \leq t(1-\alpha, df)$
  - Reject H_0 if $T > t(1-\alpha, df)$

Speaker notes

Sometimes you are only interested in changes in a positive direction. A difference only in the positive direction implies that the first population mean is larger than the second population mean. No change and changes in the negative direction are part of the null hypothesis.

If the difference in the sample, x-bar-1 minus x-bar-2 is large negative, you accept the null hypothesis. If the difference in the sample is close to zero, you accept the null hypothesis. Only if the difference in the sample, x-bar-1 minus x-bar-2 is large positive would you reject the null hypothesis.

Critical value for a one tailed test, 1



Simon, 2024-10-06

Speaker notes

The area to the right of the critical value is α and the area to the left is 1-α.

One-tailed tests, 2

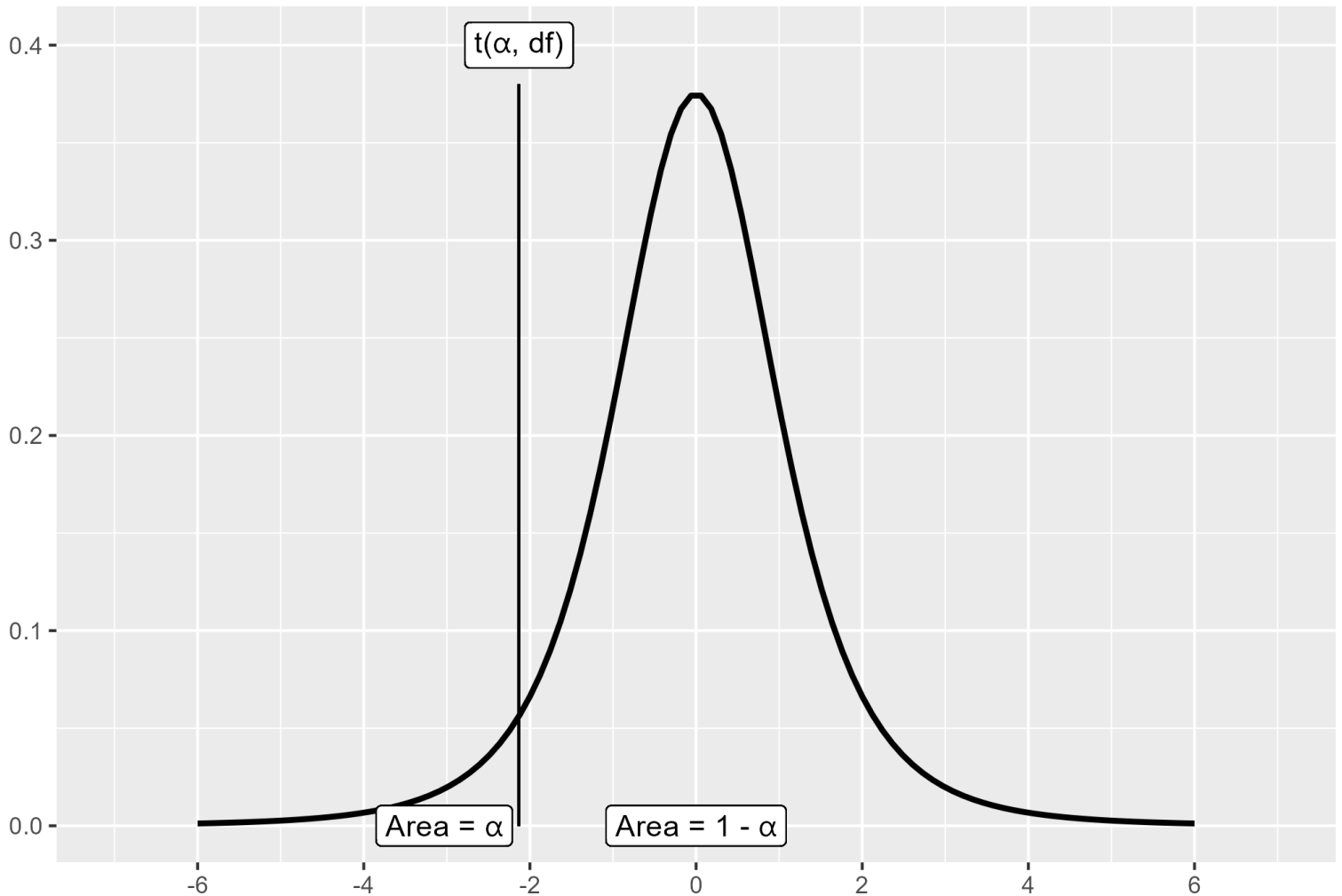
- $H_0: \ \mu_1 - \mu_2 \geq 0$
- $H_1: \ \mu_1 - \mu_2 < 0$
  - Reject H_0 if $\bar{X}_1 - \bar{X}_2$ is large negative
  - Accept H_0 if $\bar{X}_1 - \bar{X}_2$ is close to zero
  - Accept H_0 if $\bar{X}_1 - \bar{X}_2$ is large positive
- More precisely,
  - Accept H_0 if $T \geq t(\alpha, df)$
  - Reject H_0 if $T < t(\alpha, df)$

Speaker notes

There is a second one-sided hypothesis. This tests the hypothesis that the difference is negative, or equivalently that the first population mean is smaller than the second population mean.

If the difference in the sample, x-bar-1 minus x-bar-2 is large negative, you should reject the null hypothesis. If the difference in the sample is close to zero or if it is large positive, you should accept the null hypothesis.

Critical value for the one-tailed test, 2



Simon, 2024-10-06

Speaker notes

The area to the left is α and the area to the right is 1-α.

Calculation of the p-value

- For testing $H_1 : \mu_1 - \mu_2 \neq 0$
  - p-value = $2P[t(n_1 + n_2 - 2) > |T|]$
- For testing $H_1 : \mu_1 - \mu_2 > 0$
  - p-value = $P[t(n_1 + n_2 - 2) > T]$
- For testing $H_1 : \mu_1 - \mu_2 < 0$
  - p-value = $P[t(n_1 + n_2 - 2) < T]$
- For any of these hypothesis,
  - Accept H_0 if p-value > α

Speaker notes

The calculation of the p-value changes slightly if you have a one-sided hypothesis. But the decision rule is always the same, accept the null hypothesis if the p-value is large, larger than alpha.

R code for one-sided hypotheses

- For testing $H_1: \mu_1 - \mu_2 \neq 0$
  - `alternative="two-sided"`
- For testing $H_1: \mu_1 - \mu_2 > 0$
  - `alternative="greater"`
- For testing $H_1: \mu_1 - \mu_2 < 0$
  - `alternative="less"`

Speaker notes

The t.test function in R has an argument, alternative. The default is “two-sided”. Change this to “greater” or “less” to produce the appropriate one-sided hypothesis.

Break #2

- What you have learned
  - One-tailed tests
- What’s coming next
  - Checking assumptions

Equivalence of the t-test and linear regression

```
m1 <- t.test(
  fbsway ~ age,
  data=sway,
  var.equal=TRUE)
m2 <- lm(
  fbsway ~ age,
  data=sway)
```

Output from t.test and lm

```
1 m1
```

```
    Two Sample t-test

data:  fbsway by age
t = 2.2044, df = 15, p-value = 0.04353
alternative hypothesis: true difference in means between group Elderly and group
Young is not equal to 0
95 percent confidence interval:
  0.2715453 16.1451214
sample estimates:
mean in group Elderly   mean in group Young
             26.33333              18.12500
```

```
1 m2
```

```
Call:
lm(formula = fbsway ~ age, data = sway)

Coefficients:
```

Output from t.test and tidy

```
1 m1
```

```
    Two Sample t-test

data:  fbsway by age
t = 2.2044, df = 15, p-value = 0.04353
alternative hypothesis: true difference in means between group Elderly and group
Young is not equal to 0
95 percent confidence interval:
  0.2715453 16.1451214
sample estimates:
mean in group Elderly   mean in group Young
             26.33333              18.12500
```

```
1 tidy(m2)
```

```
# A tibble: 2 × 5
  term        estimate std.error statistic      p.value
  <chr>          <dbl>     <dbl>     <dbl>        <dbl>
1 (Intercept)    26.3       2.55     10.3  0.0000000334
2 ageYoung       -8.21      3.72     -2.20 0.0435
```

Output from t.test and confint

```
1 m1
```

```
    Two Sample t-test

data:  fbsway by age
t = 2.2044, df = 15, p-value = 0.04353
alternative hypothesis: true difference in means between group Elderly and group
Young is not equal to 0
95 percent confidence interval:
  0.2715453 16.1451214
sample estimates:
mean in group Elderly   mean in group Young
             26.33333              18.12500
```

```
1 confint(m2)
```

```
                2.5 %     97.5 %
(Intercept)  20.88874 31.7779250
ageYoung    -16.14512 -0.2715453
```

Population model for the two-sample t-test

- Population 1
  - $X_{11}, X_{12}, \ldots, X_{1N_1}$
  - X_{1i} are independent $N(\mu_1, \sigma_1)$
- Population 2
  - $X_{21}, X_{22}, \ldots, X_{2N_2}$
  - X_{2i} are independent $N(\mu_2, \sigma_2)$
- Both populations independent from each other
- Both populations have the same standard deviation

Speaker notes

Here is the population model for the two-sample t-test.

Violations of assumptions, linear regression

- Nonlinearity
- Non-normality
- Heterogeneity
- Lack of independence

Violations of assumptions, t-test

- ~~Nonlinearity~~
- Non-normality
- Heterogeneity
- Lack of independence
  - Within groups
  - Between groups

Speaker notes

There are several possible violations of the population model.

Checking for independence

- Qualitative assessments only
  - Clustering
  - Geographic proximity

Speaker notes

The independence needs to be evaluated qualitatively. Look for features of the data such as clustering or geographic proximity, especially for infectious diseases.

Checking for non-normality

- Boxplot
  - Might be sufficient by itself
- Non-normality is less concerning with large sample sizes
- Residual analysis
  - Normal probability plot
  - Histogram
  - Always residuals, never the original data

Speaker notes

The boxplot is a great way to check for non-normality. Look for skewness and/or outliers. Other types of non-normality, such as a bimodal distribution or a light-tailed distribution are not a serious problem. Also, non-normality is less of a concern for large sample sizes.

You really don’t need to look at residuals, but if you do, compute them the same way as in linear regression. Always examine the residuals and not the original data. If the two groups have different means (which is something that you actually hypothesize might happen), then any analysis of the original data will show a bimodal pattern.

Checking for heterogeneity

- Boxplot
- Compare group standard deviations
  - 3 to 1 or higher ratio
- Predicted values versus residuals
  - Look for a fanning out pattern
- If heterogeneity is found
  - Is the data unbalanced ($n_1 \neq n_2$)

Speaker notes

The boxplot is also a simple way to check for heterogeneity. Is one box a lot bigger than the other?

You should also compute means and standard deviations for each group. This is a routine that I always recommend. Wade in from the shallow end of the pool.

If one standard deviation is a lot larger than another, you may have problems with heterogeneity. Don’t worry about small disparities. Look for a three fold change or greater between the larger standard deviation and the smaller standard deviation.

Unbalanced data, where you have more observations in one group than the other is not a violation of assumptions, but it does tend to exacerbate problems with heterogeneity if it exists.

Consequences of violations

- Loss of control of Type I error rate
  - Two possibilities
    - Liberal, actual alpha much larger than 0.05
    - Conservative, actual alpha much smaller than 0.05
- Possible increase of Type II error rate = loss of power
  - Especially if there are outliers
- Confidence intervals too wide or too narrow

Speaker notes

What does a violation of assumptions mean from a practical perspective? First, it means that you lose control of the Type I error rate. It can change liberally, meaning that a test you intended to have a Type I error rate of 0.05 actually has a much larger Type I error rate. It can change conservatively, meaning that a test that you intended to have a Type I error rate of 0.05 actually might have a much smaller Type I error rate.

Now you might think that being conservative sounds pretty good. Why wouldn't you mind having a smaller Type I error rate? It turns out that you are robbing Peter to pay Paul. What you gain in a reduction of the Type I error rate gets taken away by an increase in the Type II error rate. This is equivalent to a loss in power.

The problem with increased Type II error rate and a loss of power is especially likely if your data has serious outliers.

Break #3

- What you have learned
  - Checking assumptions
- What’s coming next
  - Alternative tests

Alternative tests

- If you have heterogeneity
  - Welch's test
- If you have non-normality
  - Mann-Whitney-Wilcoxon test
- If you have both:
  - Log transformation
- If you have lack of independence
  - Random effects models
    - Beyond the scope of this class

Speaker notes

There are several approaches that you should consider if you have trouble with the assumptions of the two-sample t-test.

Welch’s test is a simple modification to the test if you have unequal variances. The Wilcoxon-Mann-Whitney test is a non-parametric test that does not require the assumption of normality.

If you have both heterogeneity and non-normality, and it has to be specific type of heterogeneity and non-normality, then a log transformation is good. It stretches the small values and squeezes the large values.

Lack of independence typically requires a fairly complex approach like a random effects model that is well beyond the scope of this class. This model is covered in quite a bit of detail by Anlin Cheng in MEDB 5503, Applied Biostatistics III.

Welch's test, 1

- $T = \frac{\bar{X}_1 - \bar{X}_2}{se}$
- se = standard error changes slightly
  - $se = \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}$
- df = degrees of freedom changes slightly
  - $df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{(n_1-1)n_1^2} + \frac{s_2^4}{(n_2-1)n_2^2}}$
    - Also known as Satterthwaite's approximation

Speaker notes

Welch’s test makes minor modification to the standard error and the degrees of freedom. The formula for the degrees of freedom looks difficult, but it is just involves addition, division and raising to power. It is one of those formulas that has very little intuition and it is also something that makes us glad that the computer does all the work.

Welch’s test, 2

- R code
  - `var.equal=FALSE`

Speaker notes

The R code is fairly easy. Change the argument var.equal from TRUE to FALSE.

Controversy over the use of Welch's test, 1

- Some advocate using it all the time.
  - Welch's test is valid for homogeneity OR heterogeneity
- Some advocate using a preliminary hypothesis
  - $H_0 : \ \sigma_1 = \sigma_2$
    - Levene's test
    - Bartlett's test
    - Brown-Forsythe test

Controversy over the use of Welch's test, 2

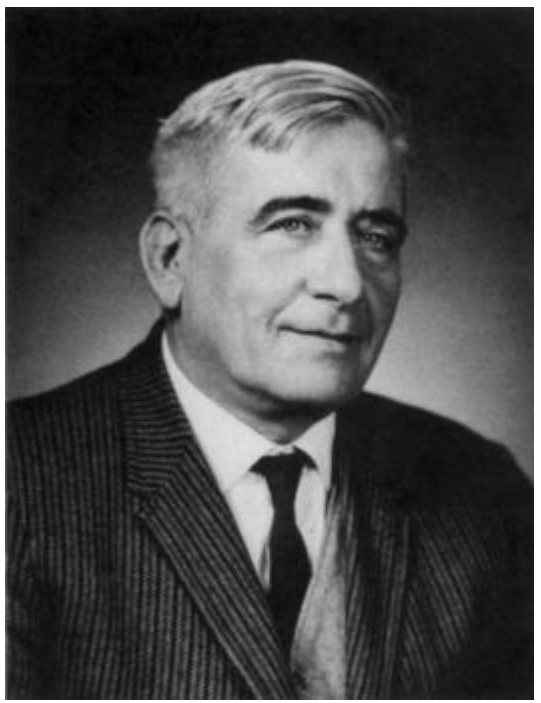
- My opinion:
  - Avoid ANY test of heterogeneity
  - Make decision based on prior experience

Speaker notes

Tests of heterogeneity are worthless. All are excessively sensitive to the normality assumption (some more than others). Worse, for very large sample sizes, these tests tend to conclude that you have heterogeneity even for trivial changes in variation from one group to another.

I do not use Welch’s test, unless there are strong a priori reasons to believe that there is heterogeneity.

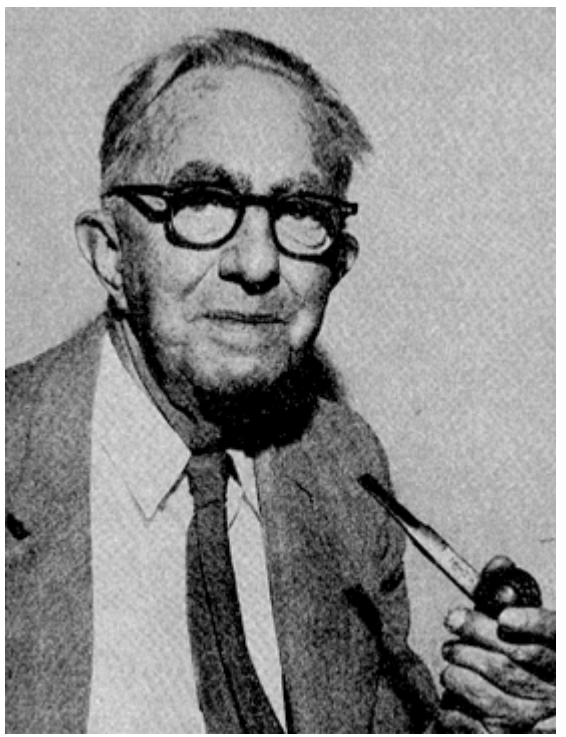
Mann-Whitney-Wilcoxon test



Henry Mann, PhD in Mathematics in 1935



Donald Ransom Whitney, PhD in Mathematics in 1946



Frank Wilcoxon, PhD in Chemistry, 1924

Speaker notes

Here are images behind the three researchers who developed the Mann-Whitney-Wilcoxon test. The websites listed below also provide nice biographies of these three people.

Henry Mann: https://math.osu.edu/about-us/history/henry-berthold-mann Donald Ransom Whitney: http://sections.maa.org/ohio/ohio_masters/whitney.pdf Frank Wilcoxon: https://en.wikipedia.org/wiki/Frank_Wilcoxon

Other names

- Wilcoxon-Mann-Whitney
- Mann-Whitney U test
- Wilcoxon rank sum test
  - Not to be confused with Wilcoxon signed rank test

Speaker notes

There are alternate names for this test. Most people actually give Frank Wilcoxon first billing. It turns out that Frank Wilcoxon published a paper in 1945 that outlined both the rank sum test and the signed rank test. The rank sum test was derived under a limited assumption of equal sample sizes in the two groups. Mann and Whitney extended this test in 1947 to make it work with unequal sample sizes.

Theory behind the Mann-Whitney-Wilcoxon test

- Combine the two groups
- Assign ranks $R(X_{ij})$
  - 1 to smallest value, 2 to second smallest, etc.
- Compute average rank, $\bar{R}$
- Compute the sum of the ranks in first group, $\Sigma R(X_{1j})$
- T = $\frac{\Sigma R(X_{1j}) - n_1 \bar{R}}{se}$
  - se = $\sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$

Alternate theory

- Count the times that X_{1j} “wins” compared to all the X_2’s
  - X_{1j} wins if it is larger than X_{2k}
  - Count ties as 0.5 (half of a “win”)
  - There are $n_1 n_2$ contests
  - U = number of wins
  - T = $\frac{U - \frac{n_1 n_2}{2}}{se}$
  - Same se as earlier slide

Hypothetical data

- X1: 34, 1695, 1193
- X2: 652, 11, 24, 16, 1543, 39

Rank sum for hypothetical data

Group	Outcome	Rank
1	34	4
1	1695	9
1	1193	7
2	652	6
2	11	1
2	24	3
2	16	2
2	1543	8
2	39	5

- $\Sigma R(X_{1j})$ = 4 + 9 + 7 = 20
- Expected sum = 3*5 = 15

Speaker notes

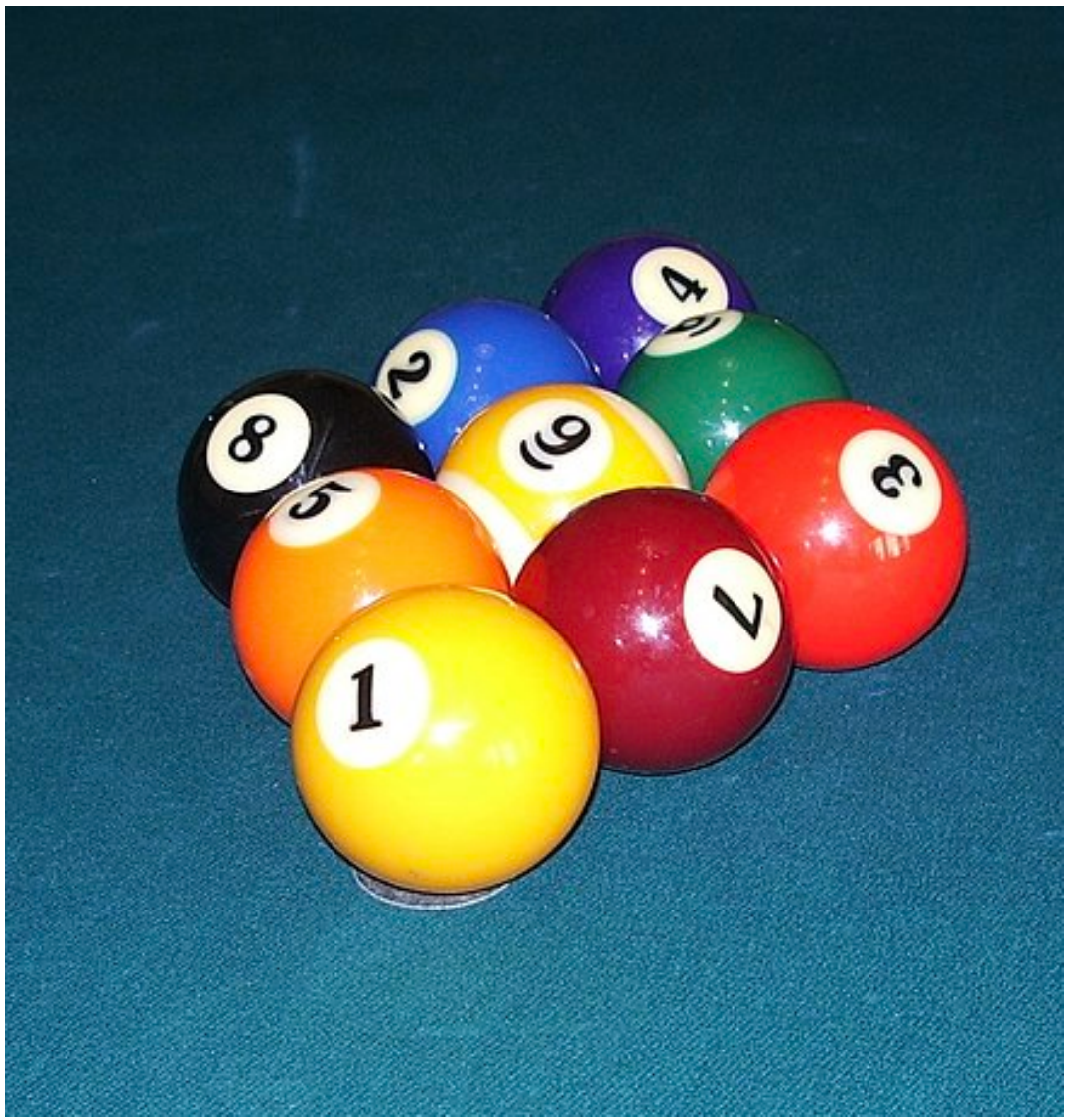
Assign the rank of 1 to the smallest value (11), a rank of 2 to the second smallest value (16),…, and a rank of 9 to the largest value (1695)

Counting “wins” for hypothetical data

	652	11	24	16	1543	39
34	Loss	Win	Win	Win	Loss	Loss
1695	Win	Win	Win	Win	Win	Win
1193	Win	Win	Win	Win	Loss	Win

- U = 14 (out of 18 possible)
- Expected wins = 9

The probability basis for ranking



Speaker notes

You randomly draw three balls from an urn containing balls numbered 1 through 9. What are the chances that you draw the a particular combination of these three balls?

List all the possible outcomes, 1

```
1,2,3  1,2,4  1,2,5  1,2,6  1,2,7  1,2,8  1,2,9
1,3,4  1,3,5  1,3,6  1,3,7  1,3,8  1,3,9  1,4,5
1,4,6  1,4,7  1,4,8  1,4,9  1,5,6  1,5,7  1,5,8
1,5,9  1,6,7  1,6,8  1,6,9  1,7,8  1,7,9  1,8,9
2,3,4  2,3,5  2,3,6  2,3,7  2,3,8  2,3,9  2,4,5
2,4,6  2,4,7  2,4,8  2,4,9  2,5,6  2,5,7  2,5,8
2,5,9  2,6,7  2,6,8  2,6,9  2,7,8  2,7,9  2,8,9
3,4,5  3,4,6  3,4,7  3,4,8  3,4,9  3,5,6  3,5,7
3,5,8  3,5,9  3,6,7  3,6,8  3,6,9  3,7,8  3,7,9
3,8,9  4,5,6  4,5,7  4,5,8  4,5,9  4,6,7  4,6,8
4,6,9  4,7,8  4,7,9  4,8,9  5,6,7  5,6,8  5,6,9
5,7,8  5,7,9  5,8,9  6,7,8  6,7,9  6,8,9  7,8,9
```

Speaker notes

Here are all the possible outcomes. There are 84 outcomes and each has probability 1/84.

List all the possible outcomes, 2

```
 6: 1,2,3
 7: 1,2,4
 8: 1,2,5  1,3,4
 9: 1,2,6  1,3,5  2,3,4
10: 1,2,7  1,3,6  1,4,5  2,3,5
11: 1,2,8  1,3,7  1,4,6  2,3,6  2,4,5
12: 1,2,9  1,3,8  1,4,7  1,5,6  2,3,7  2,4,6  3,4,5
13: 1,3,9  1,4,8  1,5,7  2,3,8  2,4,7  2,5,6  3,4,6
14: 1,4,9  1,5,8  1,6,7  2,3,9  2,4,8  2,5,7  3,4,7  3,5,6
```

List all the possible outcomes, 3

```
15: 1,5,9  1,6,8  2,4,9  2,5,8  2,6,7  3,4,8  3,5,7  4,5,6
16: 1,6,9  1,7,8  2,5,9  2,6,8  3,4,9  3,5,8  3,6,7  4,5,7
17: 1,7,9  2,6,9  2,7,8  3,5,9  3,6,8  4,5,8  4,6,7
18: 1,8,9  2,7,9  3,6,9  3,7,8  4,5,9  4,6,8  5,6,7
19: 2,8,9  3,7,9  4,6,9  4,7,8  5,6,8
20: 3,8,9  4,7,9  5,6,9  5,7,8
21: 4,8,9  5,7,9  6,7,8
22: 5,8,9  6,7,9
23: 6,8,9
24: 7,8,9
```

Controversy over the use of nonparametric tests

- Unclear what your hypothesis is
  - Inequality in population means
  - Inequality in population medians
  - $P[X > Y] \neq \frac{1}{2}$
- Loss of power
  - Multiply t-test sample size by pi/3.

Log transformation

- Arithmetic mean of original data
  - $\frac{1}{n}\Sigma X_i$
- Arithmetic mean of log transformed data
  - $\frac{1}{n}\Sigma log(X_i)$
  - $\frac{1}{n}log(X_1 \times X_2 \times \ldots \times X_n)$
- Arithmetic mean of log transformed data back transformed
  - $(X_1 \times X_2 \times \ \ldots \ \ \times X_n)^{\frac{1}{n}}$
  - This is known as the geometric mean

Break #4

- What you have learned
  - Alternative tests
- What’s coming next
  - R code for alternative tests

simon-5501-09-demo, part 1

Break #5

- What you have learned
  - R code for alternative tests
- What’s coming next
  - Paired data

Paired data

- Sometimes you collect data in pairs
  - Drug for left arm, placebo for right arm
  - Measure patient before intervention and after
  - Each exposed patient matched on age/sex with control

Speaker notes

There are lots of situations where the two groups in a research study are paired together. In this case, the test changes

Calculate differences

- $D_i = X_{1i} - X_{2i}$
  - $\bar{D} = \frac{1}{n}\Sigma D_i$
  - $S_D = \sqrt{\frac{1}{n-1}\Sigma(D_i - \bar{D})^2}$

Test of hypothesis

- $H_0 :\ \mu_D = 0$
- $H_1 :\ \mu_D \neq 0$
  - T = $\frac{\bar{D}}{se}$
    - $se = \frac{S_D}{\sqrt{n}}$
- Accept H_0 if T is close to zero.

Confidence interval

- $\bar{D} \pm t(1-\alpha/2, df)se$
  - Range of plausible values for μ_D

Violations of the assumptions of paired t-test

- ~~Non-linearity~~
- ~~Heterogeneity~~
- Normality
- Independence
  - Between pairs, not within pairs

Break #6

- What you have learned
  - Paired data
- What's coming next
  - R code for paired data

simon-5501-09-demo, part 1

Break #7

- What you have learned
  - R code for paired data
- What’s coming next
  - Your homework

simon-5501-09-directions.md

Summary

- What you have learned
  - Restructuring your data
  - One-tailed tests
  - Checking assumptions
  - Alternative tests
  - R code for alternative tests
  - Paired data
  - R code for paired data
  - Your homework